Robustness of the weighted BPX factor for $M + \sqrt{\beta}K$

Proof of Assumption 3.4 of the preconditioning paper (draft for checking)

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Status. Steps marked **[proved]** are proved here completely. Steps marked **[cited]** use classical multilevel results, stated precisely as Lemmas 4 and 5; their exact sources and constants still have to be checked. Notation: $\nu := \sqrt{\beta} > 0$ (the paper uses ε for the solver accuracy).

1 Setting

Let $\Omega = (0,1)^d$, $h_\ell = 2^{-\ell}$, and let $V_\ell \subset H_0^1(\Omega)$ be the Q_1 finite element space on the uniform grid of mesh size h_ℓ, $\ell = 1, \dots, L$. The spaces are nested, $V_1 \subset \cdots \subset V_L =: V$. For $u \in V_\ell$ let $y \in \mathbb{R}^{n_\ell}$ be its vector of nodal values. Define

$$a(u,v) = (u,v) + \nu(\nabla u, \nabla v), \qquad t_\ell := \nu h_\ell^{-2} = \nu 4^\ell, \qquad \mu_\ell := 1 + t_\ell.$$

The matrix of a in the nodal basis of V_ℓ is $A_\ell = M_\ell + \nu K_\ell$; on the finest level $A_L = A_\beta$ of the paper.

BPX factor. P_ℓ maps nodal values on level ℓ to nodal values on level L of the same function (the natural embedding $V_\ell \subset V_L$; $P_L = I$). $\operatorname{diag}(A_\ell) = \delta_\ell I$ is constant on the uniform grid. With weights $\omega_\ell > 0$,

$$C = \sum_{\ell=1}^{L} \omega_\ell^2 P_\ell P_\ell^T, \qquad \omega_\ell^2 = \frac{\tilde\theta_\ell}{\delta_\ell}.$$

The paper uses $\theta_\ell = \min(1, t_\ell)^s$ with $s \in [0,1]$. Here we use the *normalized* weights

$$\tilde\theta_\ell = \frac{\theta_\ell}{\theta_L} = \Big(\frac{\min(1,t_\ell)}{\min(1,t_L)}\Big)^s.$$

Since $\tilde\theta_\ell$ differs from θ_ℓ by the factor θ_L^{-1}, which does not depend on ℓ, C only changes by a positive scalar. So $\kappa(CA_\beta)$ is the same for both choices. The *interval* $[c_0, c_1]$ below, however, is independent of β only for the normalized weights (relevant for the heat problem, Corollary 12).

Claim (Assumption 3.4). For $s \in (0,1]$ there are $0 < c_0 \le c_1$, depending only on d and s, with $\operatorname{spec}(CA_\beta) \subset [c_0, c_1]$ for all $L \ge 1$ and $\nu > 0$.

2 Reduction to a decomposition problem

Lemma 1 (Lions; Xu–Zikatanov)**.** ***[proved]*** *Let $C = \sum_k R_k D_k^{-1} R_k^T$ with D_k symmetric positive definite and $\sum_k \operatorname{range}(R_k) = \mathbb{R}^n$. Then C is SPD and, for every $x \in \mathbb{R}^n$,*

$$x^T C^{-1} x = \min\Big\{ \sum_k y_k^T D_k y_k \;:\; \sum_k R_k y_k = x \Big\}.$$

Proof. C is SPD because $x^TCx = \sum_k |D_k^{-1/2}R_k^Tx|^2 = 0$ forces $R_k^Tx = 0$ for all k, hence $x \perp \sum_k \operatorname{range} R_k = \mathbb{R}^n$. For any decomposition $x = \sum_k R_k y_k$, by Cauchy–Schwarz,

$$x^TC^{-1}x = \sum_k y_k^T R_k^T C^{-1}x \le \Big(\sum_k y_k^T D_k y_k\Big)^{1/2}\Big(\sum_k x^TC^{-1}R_kD_k^{-1}R_k^TC^{-1}x\Big)^{1/2},$$

and the last sum equals $x^TC^{-1}CC^{-1}x = x^TC^{-1}x$. Hence $x^TC^{-1}x \le \sum_k y_k^T D_k y_k$. Equality holds for $y_k = D_k^{-1}R_k^TC^{-1}x$, which is a decomposition of x. □

Apply this with $R_\ell = P_\ell$ and $D_\ell = \omega_\ell^{-2}I = (\delta_\ell/\tilde\theta_\ell)I$; $P_L = I$ makes the ranges span. Let x be the nodal vector of $u \in V$ and y_ℓ that of $u_\ell \in V_\ell$; then $\sum_\ell P_\ell y_\ell = x$ means $\sum_\ell u_\ell = u$, and $x^TA_\beta x = a(u,u)$. So

$$x^TC^{-1}x = \min_{u=\sum_\ell u_\ell,\ u_\ell\in V_\ell} \sum_\ell \frac{\delta_\ell}{\tilde\theta_\ell}|y_\ell|^2. \tag{1}$$

Since $\operatorname{spec}(CA_\beta) \subset [c_0, c_1] \iff c_1^{-1}A_\beta \preceq C^{-1} \preceq c_0^{-1}A_\beta$, the claim is equivalent to

$$c_1^{-1}\,a(u,u) \;\le\; x^TC^{-1}x \;\le\; c_0^{-1}\,a(u,u) \qquad \text{for all } u \in V. \tag{2}$$

3 From nodal coefficients to L^2 norms

Lemma 2. ***[proved]*** *For $v \in V_\ell$ with nodal vector y,*

$$\Big(\frac{h_\ell}{3}\Big)^d|y|^2 < \|v\|^2 < h_\ell^d|y|^2, \qquad \delta_\ell = \Big(\frac{2h_\ell}{3}\Big)^d\big(1+3d\,t_\ell\big) \in \Big[\Big(\frac{2h_\ell}{3}\Big)^d\mu_\ell,\ 3d\Big(\frac{2h_\ell}{3}\Big)^d\mu_\ell\Big].$$

Proof. $\|v\|^2 = y^TM_\ell y$ with $M_\ell = M_1^{\otimes d}$, and the 1D mass matrix $\frac{h}{6}\operatorname{tridiag}(1,4,1)$ has eigenvalues $\frac{h}{6}(4+2\cos\phi_j) \in (\frac{h}{3}, h)$. The diagonal entries are $(M_\ell)_{ii} = (2h/3)^d$ and $(K_\ell)_{ii} = d\cdot\frac{2}{h}(\frac{2h}{3})^{d-1}$, so $\delta_\ell = (\frac{2h}{3})^d + \nu\frac{2d}{h}(\frac{2h}{3})^{d-1} = (\frac{2h}{3})^d(1+3d\nu h^{-2})$. Finally $\mu \le 1+3dt \le 3d\mu$ for $t \ge 0$ and $d \ge 1$. □

Write $\lambda_\ell := \mu_\ell/\tilde\theta_\ell$ and

$$N(u) := \inf_{u=\sum_\ell u_\ell,\ u_\ell\in V_\ell} \sum_\ell \lambda_\ell\|u_\ell\|^2.$$

By Lemma 2, for each term $(\frac{2}{3})^d\lambda_\ell\|u_\ell\|^2 \le \frac{\delta_\ell}{\tilde\theta_\ell}|y_\ell|^2 \le 3d\,2^d\lambda_\ell\|u_\ell\|^2$ (the upper bound uses $h_\ell^d|y_\ell|^2 < 3^d\|u_\ell\|^2$, the lower bound $\|u_\ell\|^2 < h_\ell^d|y_\ell|^2$). Taking the infimum in (1),

$$\big(\tfrac{2}{3}\big)^d N(u) \;\le\; x^TC^{-1}x \;\le\; 3d\,2^d\,N(u). \tag{3}$$

It remains to show $K_1^{-1}a(u,u) \le N(u) \le K_0\,a(u,u)$ with K_0, K_1 independent of L and ν. Then (2) holds with

$$c_0 = \frac{1}{3d\,2^dK_0}, \qquad c_1 = \big(\tfrac{3}{2}\big)^dK_1, \qquad \frac{c_1}{c_0} \le 3d\cdot 3^d\cdot K_0K_1.$$

The factor $3d\cdot 3^d$ comes only from Lemma 2, i.e. from Jacobi scaling and the nodal basis. It is the "mass-matrix floor" seen in the numerics.

4 Two regimes and the weights

Since $t_\ell = \nu 4^\ell$ increases by the factor 4 per level, there are two regimes.

- *Unresolved* ($t_L < 1$): all levels are mass dominated. Then $\tilde{\theta}_\ell = (t_\ell/t_L)^s = 4^{-s(L-\ell)}$ and $\lambda_\ell = (1+t_\ell)\, 4^{s(L-\ell)}$.
- *Resolved* ($t_L \geq 1$): let $\ell^* = \min\{\ell : t_\ell \geq 1\}$. Then $\tilde{\theta}_\ell = \theta_\ell$, and $\lambda_\ell = 1 + t_\ell \in [t_\ell, 2t_\ell]$ for $\ell \geq \ell^*$, while $\lambda_\ell = (1+t_\ell) t_\ell^{-s} \in [t_\ell^{-s}, 2t_\ell^{-s}]$ for $\ell < \ell^*$. Moreover $t_{\ell^*} < 4$ if $\ell^* > 1$.

Lemma 3. ***[proved]*** *For $s \in (0,1]$, in both regimes,*

$$\textit{(a)}\quad t_\ell \leq \lambda_\ell \ \textit{ for all } \ell, \qquad \textit{(b)}\quad S := \sum_{\ell=1}^{L} \lambda_\ell^{-1} \leq \frac{4}{3} + \frac{1}{1-4^{-s}}.$$

For $s = 0$, (a) still holds, but S is of the order of the number of mass-dominated levels: $S \approx L$ in the unresolved regime and $S \approx \ell^* \approx \log_4(1/\nu)$ in the resolved one.

Proof. (a) If $t_\ell \geq 1$ then $\lambda_\ell = 1 + t_\ell > t_\ell$. If $t_\ell < 1$ then $\lambda_\ell \geq 1 > t_\ell$, since $\tilde{\theta}_\ell \leq 1$ in both regimes. (b) Unresolved: $\lambda_\ell^{-1} \leq 4^{-s(L-\ell)}$, so $S \leq \sum_{j\geq 0} 4^{-sj} = 1/(1-4^{-s})$. Resolved: for $\ell \geq \ell^*$, $\lambda_\ell^{-1} \leq t_\ell^{-1} = t_{\ell^*}^{-1} 4^{-(\ell-\ell^*)} \leq 4^{-(\ell-\ell^*)}$, which sums to at most $4/3$. For $\ell < \ell^*$, $\lambda_\ell^{-1} \leq t_\ell^s = t_{\ell^*-1}^s 4^{-s(\ell^*-1-\ell)} < 4^{-s(\ell^*-1-\ell)}$, which sums to at most $1/(1-4^{-s})$. For $s = 0$ the bounds $\lambda_\ell^{-1} \leq 1$ on the mass-dominated levels are attained up to the factor $1 + t_\ell < 2$. □

5 Two classical ingredients

Let $Q_\ell : L^2(\Omega) \to V_\ell$ be the L^2-orthogonal projection, $Q_0 := 0$.

Lemma 4 (multilevel norm equivalence, Jackson direction)**.** ***[cited]*** *There is C_Q, depending only on d, such that for all L and all $u \in V_L$*

$$\sum_{\ell=1}^{L} h_\ell^{-2} \|(Q_\ell - Q_{\ell-1})u\|^2 \leq C_Q\, |u|_{H^1}^2.$$

Sources to check:

- Oswald, *Multilevel Finite Element Approximation*, 1994;
- Bornemann–Yserentant, Numer. Math. 64, 1993;
- Dahmen–Kunoth, Numer. Math. 63, 1992;
- Xu, SIAM Rev. 34, 1992.

Note that the easy argument via $\|(I - Q_\ell)u\| \lesssim h_\ell |u|_1$ only gives $C_Q = O(L)$; the L-independent bound is the deep part of BPX theory.

Lemma 5 (strengthened Cauchy–Schwarz and inverse inequality)**.** ***[cited]*** *There are C_S and C_I, depending only on d, such that for $k \leq \ell$, $u \in V_k$, $v \in V_\ell$,*

$$|(\nabla u, \nabla v)| \leq C_S\, 2^{-(\ell-k)/2}\, |u|_{H^1}\, h_\ell^{-1} \|v\|, \qquad |v|_{H^1}^2 \leq C_I h_\ell^{-2} \|v\|^2.$$

Sources to check: Yserentant (Numer. Math. 49, 1986), Xu (SIAM Rev. 1992), Bramble–Pasciak–Xu (Math. Comp. 1990). For Q_1 on a uniform grid the inverse inequality holds with $C_I = 4d \cdot 3^d$, by the eigenvalue bounds $y^T K_\ell y \leq 4d\, h_\ell^{d-2} |y|^2$ and Lemma 2; an elementwise argument gives a smaller constant. The strengthened Cauchy–Schwarz inequality uses that Q_1 functions are harmonic on each coarse element (every d-linear monomial is harmonic), integration by parts on coarse elements, and a boundary-strip estimate for v.

Lemma 6 (H^1 part of the upper bound)**.** ***[proved from Lemma 5]*** *For every decomposition $u = \sum_\ell u_\ell$, $u_\ell \in V_\ell$,*

$$\nu|u|_{H^1}^2 \le C_H \sum_\ell t_\ell \|u_\ell\|^2, \qquad C_H := C_I + \frac{2C_S C_I^{1/2}}{\sqrt{2}-1}.$$

Proof. Let $a_\ell := t_\ell^{1/2}\|u_\ell\| = \nu^{1/2} h_\ell^{-1}\|u_\ell\|$. Expanding, $\nu|u|_1^2 = \nu\sum_\ell |u_\ell|_1^2 + 2\nu\sum_{k<\ell}(\nabla u_k, \nabla u_\ell)$. The diagonal is at most $C_I\sum_\ell a_\ell^2$ by the inverse inequality. For $k<\ell$, Lemma 5 and the inverse inequality for u_k give $\nu|(\nabla u_k, \nabla u_\ell)| \le C_S C_I^{1/2} 2^{-(\ell-k)/2} a_k a_\ell$. With $q = 2^{-1/2}$, $\sum_{k<\ell} q^{\ell-k} a_k a_\ell \le \big(\sum_{j\ge 1} q^j\big)\sum_\ell a_\ell^2 = \frac{1}{\sqrt{2}-1}\sum_\ell a_\ell^2$ by the Schur test ($2a_k a_\ell \le a_k^2 + a_\ell^2$). □

6 The two bounds

Lemma 7 (upper bound for a)**.** ***[proved, given Lemma 5]*** *For $s \in (0,1]$ and every decomposition $u = \sum_\ell u_\ell$,*

$$a(u,u) \le K_1 \sum_\ell \lambda_\ell \|u_\ell\|^2, \qquad K_1 := \frac{4}{3} + \frac{1}{1-4^{-s}} + C_H.$$

Consequently $a(u,u) \le K_1 N(u)$.

Proof. Mass part: by Cauchy–Schwarz, $\|\sum_\ell u_\ell\| \le \sum_\ell \lambda_\ell^{-1/2}\cdot\lambda_\ell^{1/2}\|u_\ell\| \le S^{1/2}(\sum_\ell \lambda_\ell\|u_\ell\|^2)^{1/2}$, and S is bounded by Lemma 3(b). Stiffness part: Lemma 6 and $t_\ell \le \lambda_\ell$ (Lemma 3(a)). □

Lemma 8 (stable decomposition)**.** ***[proved, given Lemma 4]*** *For every $u \in V$, $N(u) \le K_0\, a(u,u)$ with $K_0 := \max(5, 2C_Q)$, for every $s \in [0,1]$.*

Proof. Unresolved regime. Take $u_L = u$ and $u_\ell = 0$ otherwise. Then $\lambda_L = 1 + t_L < 2$ and $N(u) \le 2\|u\|^2 \le 2a(u,u)$.

Resolved regime. Take $u_\ell = 0$ for $\ell < \ell^*$, $u_{\ell^*} = Q_{\ell^*}u$, and $u_\ell = (Q_\ell - Q_{\ell-1})u$ for $\ell > \ell^*$. These sum to u, since $Q_L u = u$. For $\ell > \ell^*$, $\lambda_\ell \le 2t_\ell$. For ℓ^*:

- if $\ell^* > 1$: $\lambda_{\ell^*} = 1 + t_{\ell^*} < 5$ and $\|Q_{\ell^*}u\| \le \|u\|$, so this term is at most $5\|u\|^2$;
- if $\ell^* = 1$: $\lambda_1\|Q_1 u\|^2 \le \|u\|^2 + t_1\|(Q_1 - Q_0)u\|^2$.

In both cases

$$N(u) \le 5\|u\|^2 + 2\nu\sum_{\ell\ge 1} h_\ell^{-2}\|(Q_\ell - Q_{\ell-1})u\|^2 \le 5\|u\|^2 + 2C_Q\,\nu|u|_1^2 \le K_0\, a(u,u)$$

by Lemma 4. □

7 Result

Theorem 9. *Let $s \in (0,1]$ and use the normalized weights $\tilde\theta_\ell$. For all $L \ge 1$ and $\beta > 0$,*

$$\operatorname{spec}(CA_\beta) \subset [c_0, c_1], \qquad c_0 = \frac{1}{3d\,2^d\max(5,2C_Q)}, \qquad c_1 = \big(\tfrac{3}{2}\big)^d\Big(\frac{4}{3} + \frac{1}{1-4^{-s}} + C_H\Big).$$

In particular $\kappa(CA_\beta) \le 3d\cdot 3^d \max(5,2C_Q)\big(\frac{4}{3} + \frac{1}{1-4^{-s}} + C_H\big)$, independently of h and β. The same condition number bound holds for the unnormalized weights θ_ℓ of the paper.

Proof. Combine (3) with Lemmas 7 and 8. The last sentence follows from the scale invariance of κ. □

Remark 10 ($s = 0$, plain BPX). Lemma 8 still holds, but in Lemma 7 the mass part contributes S, which is of the order of the number of mass-dominated levels. The resulting bound grows like $\log(1/\beta)$ in the resolved regime and like L in the unresolved regime. This matches the observed degradation of plain BPX. It is also consistent with the L-dependence in Yang–Yu–Zhang (arXiv:2610.06987), whose weights do not damp mass-dominated levels (to be checked against their proof).

Remark 11 (sharpness). The constants are far from sharp: for $d = 3$ the factor $3d \cdot 3^d = 243$ alone exceeds the observed $\kappa(CA_\beta) \leq 20.5$. The proof explains the structure (geometric damping makes the mass part summable; the floor comes from Jacobi scaling), not the size of the constants.

Corollary 12 (time-periodic heat problem)**.** *For frequency k the paper's operator is $A_k = r_k M + \nu K = r_k(M + \nu_k K)$ with $\nu_k = \nu/r_k$. Using in R_k the diagonal of A_k and the normalized weights for ν_k, we get $C_k A_k = C(\nu_k) A_{\nu_k}$ with $C(\nu_k)$ the factor of Theorem 9 for parameter ν_k. Hence* $\operatorname{spec}(C_k A_k) \subset [c_0, c_1]$ *with the* same c_0, c_1 *for all k. This is the hypothesis of the paper's heat theorem.*

Proof. $\operatorname{diag}(A_{k,\ell}) = r_k \operatorname{diag}(M_\ell + \nu_k K_\ell)$, so $\omega_{\ell,k}^2 = r_k^{-1} \omega_\ell(\nu_k)^2$ and $C_k = r_k^{-1} C(\nu_k)$. Then $C_k A_k = C(\nu_k)(M + \nu_k K)$, and Theorem 9 applies with ν_k; its interval does not depend on ν. □